EE 105 · Module 0
Lab: Arduino Setup

September 1, 2026

Lab Check-in

ee105.org/checkin

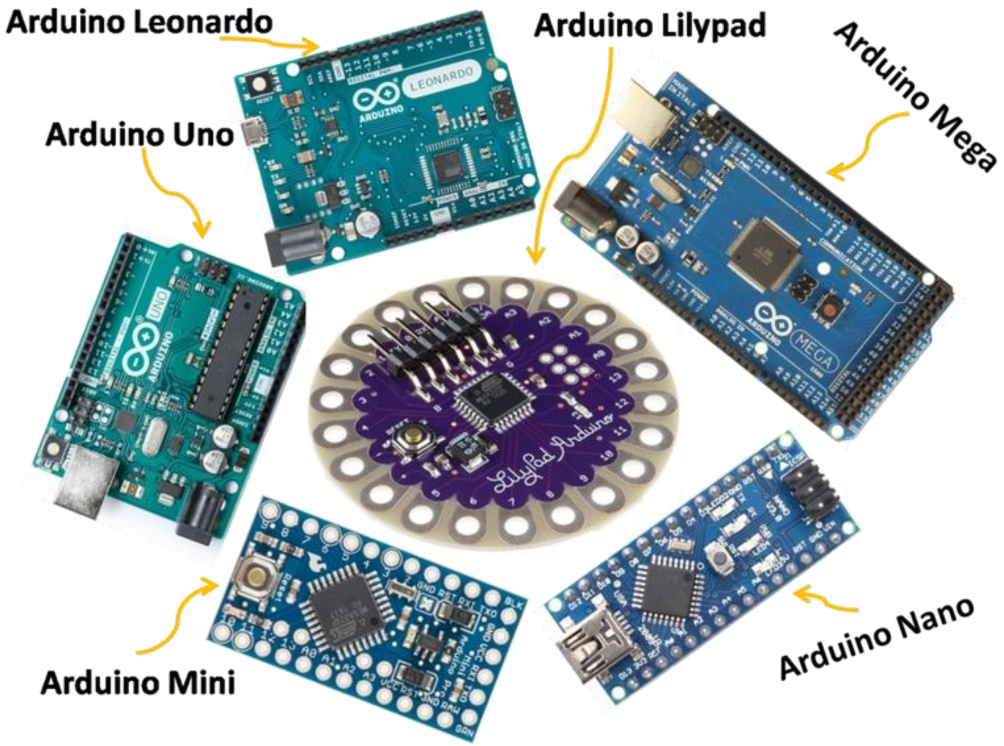
Scan with your phone and answer the quick questions — your responses steer today's lab.
The page is open only during lab hours.

What is Arduino?

- Microcontroller Development Board
- Open-source hardware + software.
- Simple programming with Arduino IDE.
- Used for rapid prototyping.

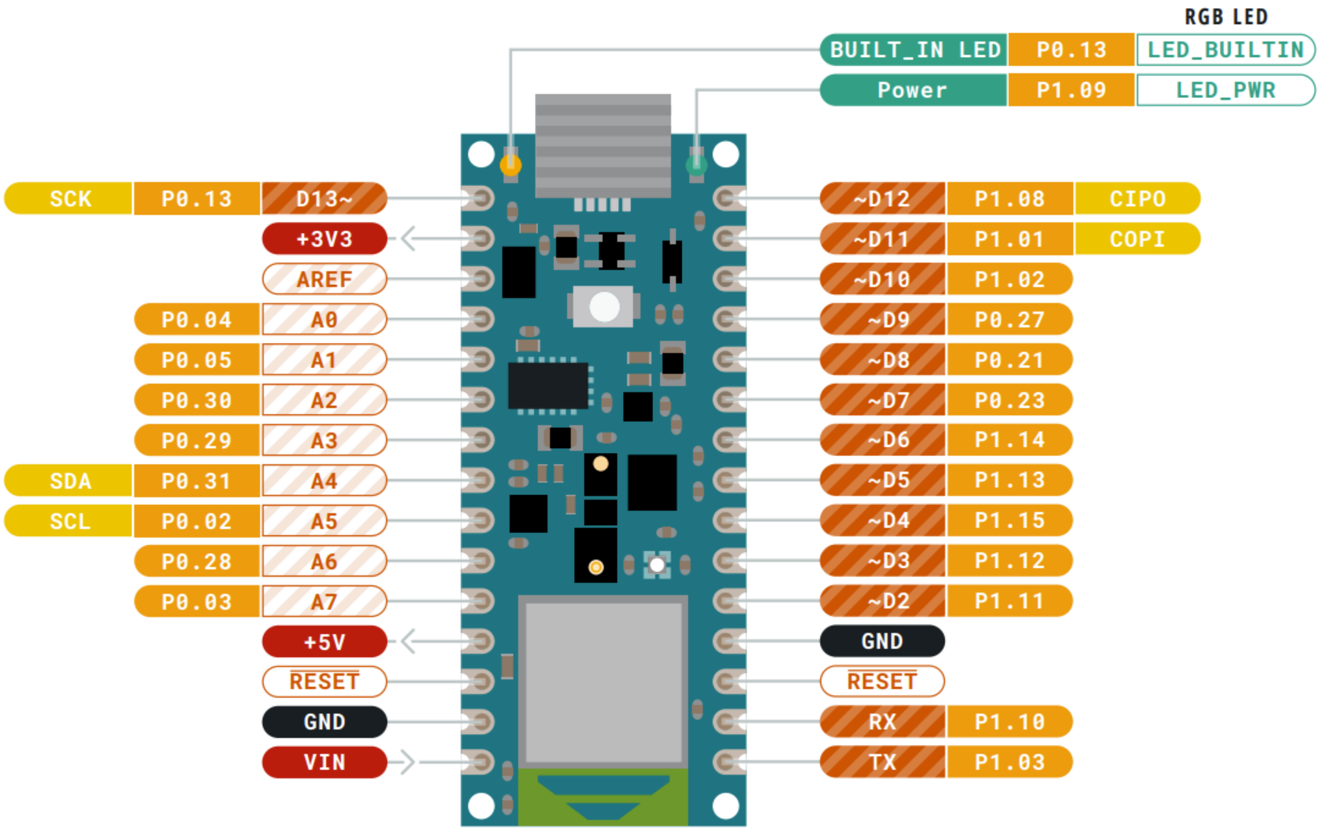
Why use Arduino?

- Low cost and widely available.
- Large community and libraries.
- Easy to connect sensors/actuators.
- Good entry point into embedded systems.



Different types of Arduino Boards

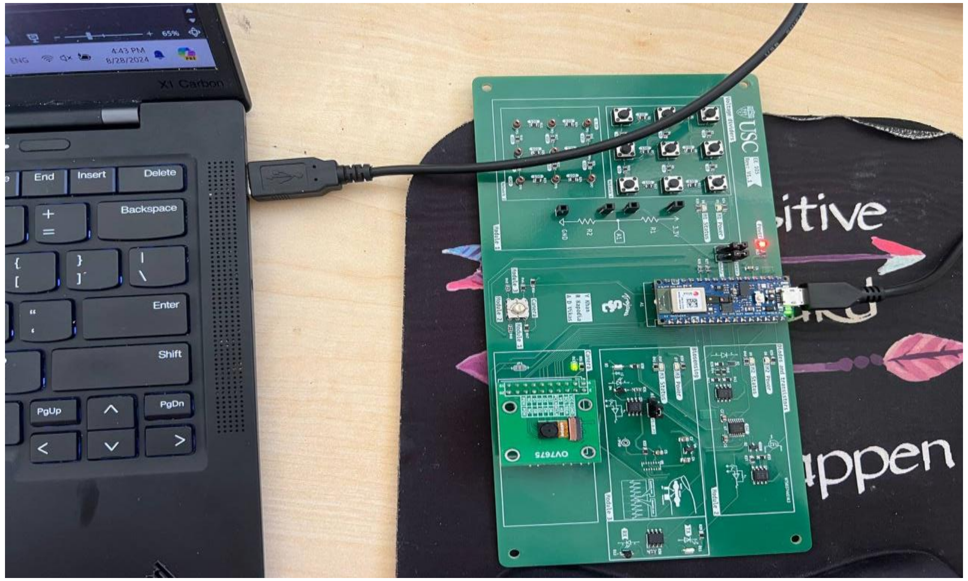
Arduino Nano 33 BLE



Installing the Arduino IDE

1. Connect the USB cable to Arduino

2. Download the latest release of the Arduino IDE here
[support.arduino.cc → Download and install Arduino IDE]()



Scan to download

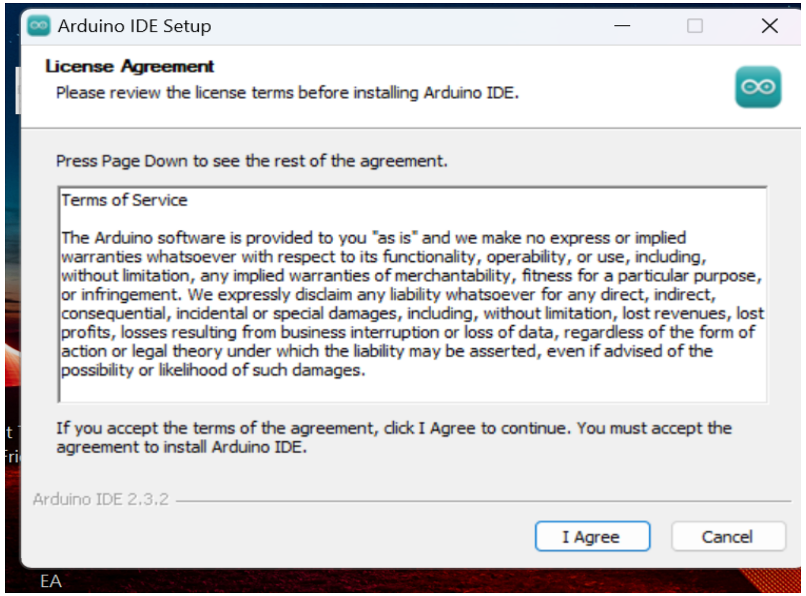
Run the Installer

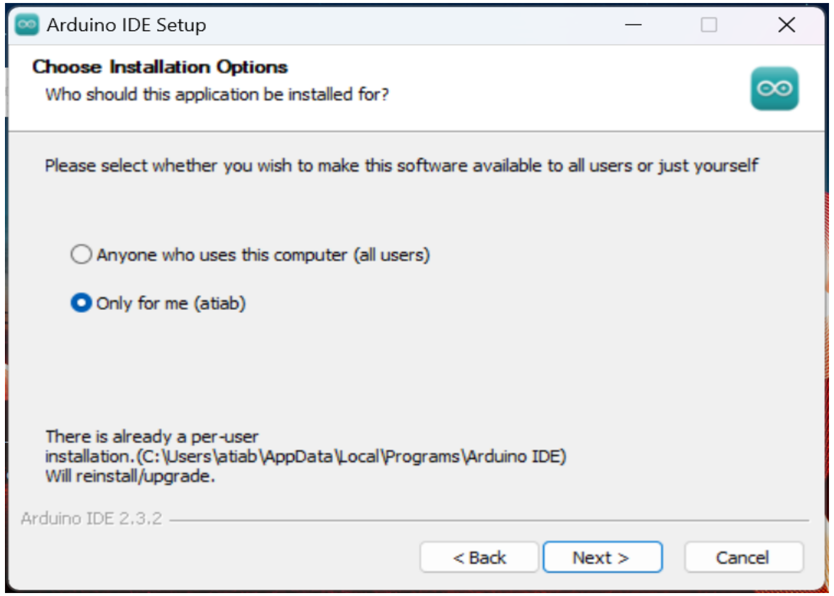
1

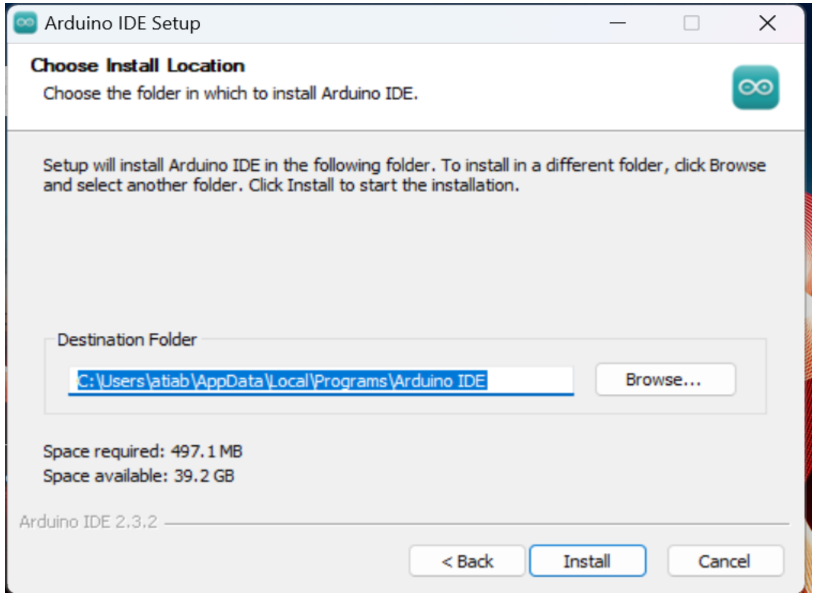
2

3

Windows



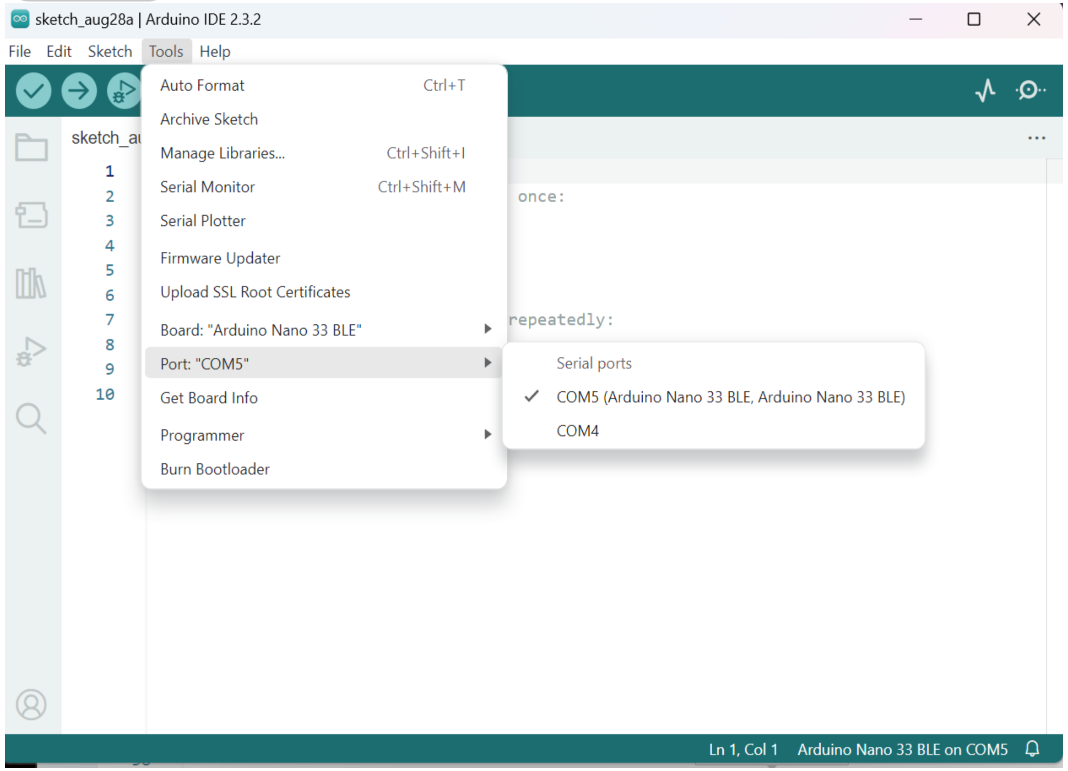




Select Board and Port

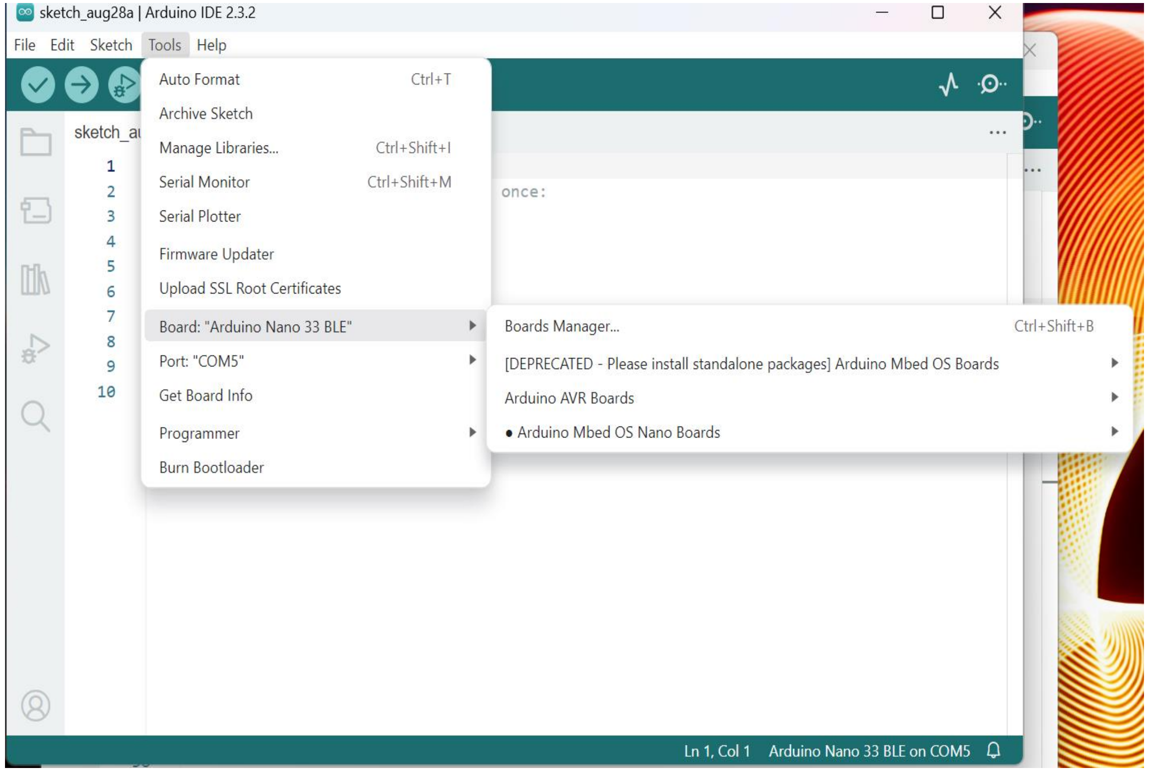
1. Identify the COM port that Arduino is connected

Windows



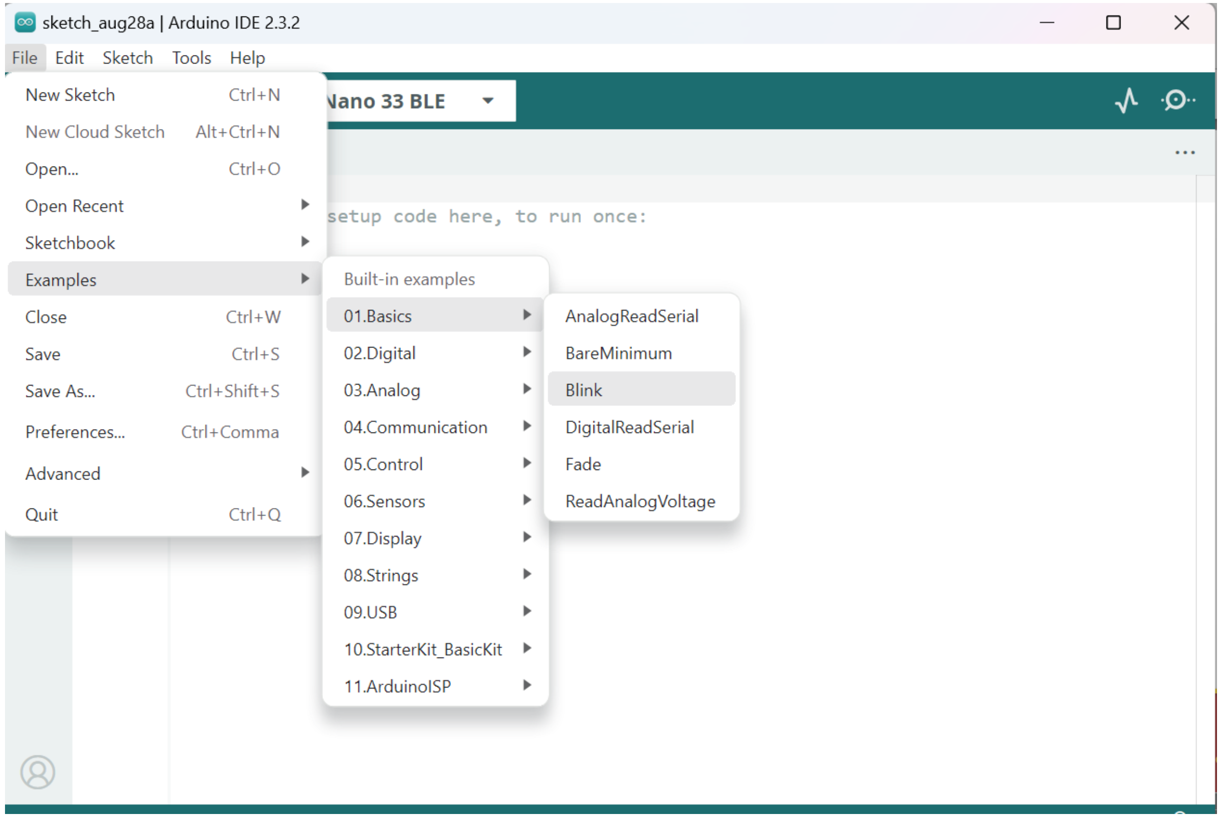
2. Ensure the Mbed OS Nano Boards package is installed

`Tools -> Board: -> Boards Manager... -> Install "Arduino Mbed OS Nano Boards"`



Your First Sketch: Blink

Windows




```
void setup() {
  // initialize digital pin LED_BUILTIN as an
output.
  pinMode(LED_BUILTIN, OUTPUT);
}

// loop() repeats forever
void loop() {
  digitalWrite(LED_BUILTIN, HIGH);  // LED on
  delay(1000);   // wait for a second
  digitalWrite(LED_BUILTIN, LOW);   // LED off
  delay(1000);   // wait for a second
}
```

Blink, Refactored

Another way of coding

```
// Pin number for the built-in LED
const int ledPin = 13;

void setup() {
  // initialize digital pin 13 as an output.
  pinMode(ledPin, OUTPUT);
}

void loop() {
  digitalWrite(ledPin, HIGH);   // turn the LED on
  delay(1000);    // wait for a second
  digitalWrite(ledPin, LOW);    // turn the LED off
  delay(1000);    // wait for a second
}
```

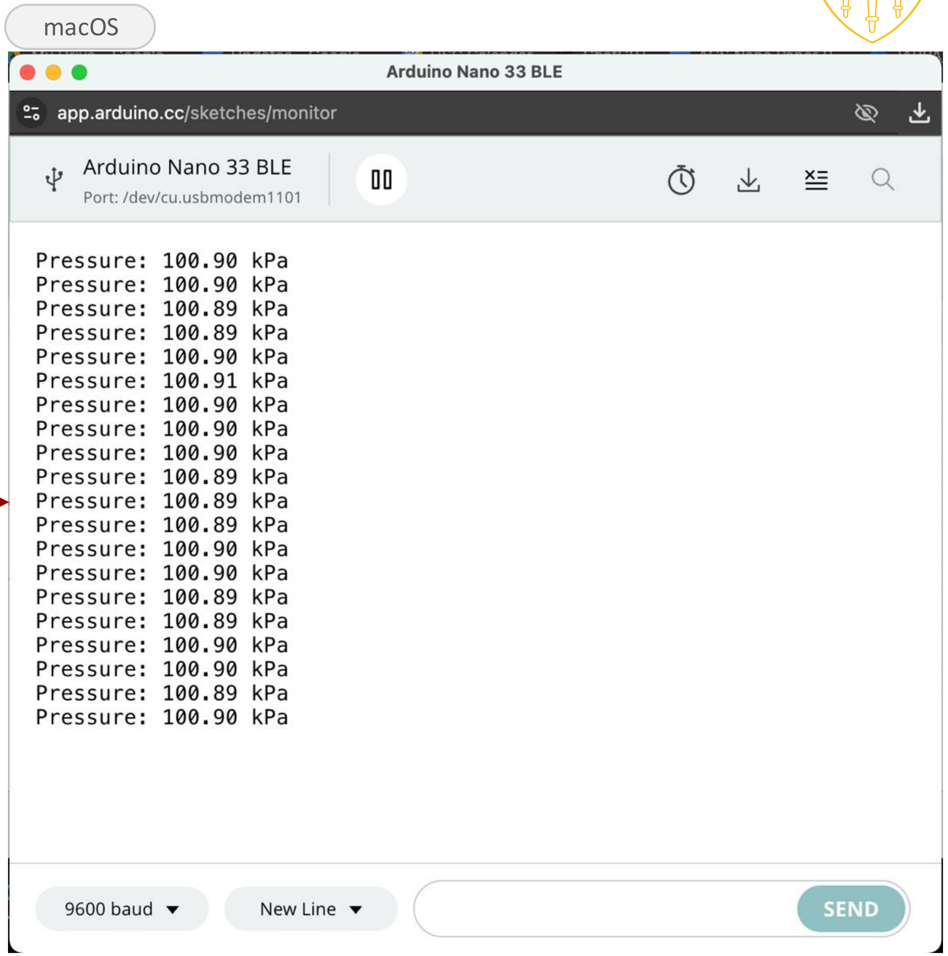
Notice: the frequency of the LED is **0.5 Hz** — the two highlighted delay(1000) calls set the 2-second period.

Using the Serial Monitor

- Testing pressure sensor: Open the sketch in **‘pressure_mesurement’**

[app.arduino.cc → pressure-sensor test sketch]

Your serial monitor output should look like this



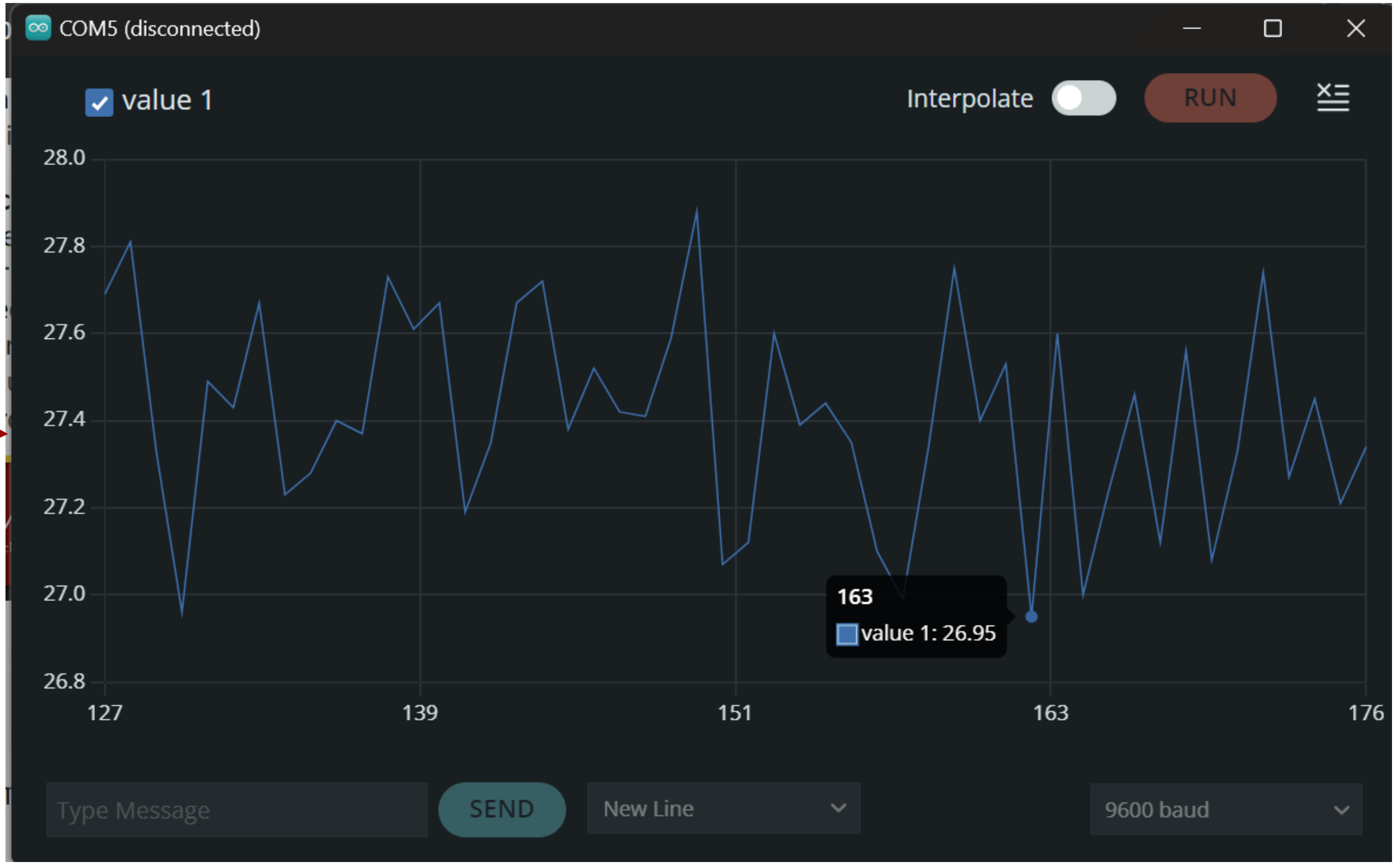
Scan for the sketch

Using the Serial Plotter

- Just like Serial monitor
- Plots values with time in x axis
- Use Serial.println() instead of Serial.print()

Your serial plotter output should look like this

Windows · dark mode



The Arduino Nano 33 BLE is equipped with several built-in sensors, making it a powerful tool for various applications without needing additional hardware.

•**APDS9960**: Detects ambient light, color, proximity, and gestures.

•**LPS22HB**: Measures atmospheric pressure

•**LSM9DS1**: Tracks acceleration, rotation, and magnetic field orientation.

•**PDM Microphone**: Captures and analyzes sound signals.

•**RGB LED**: Provides visual feedback with controllable colors.

What the Sensors Can Do — Light & Gesture

Color Detection

The APDS9960 measures the red, green, and blue light reflected from an object — the web app's Colorimeter uses it to identify colors.

Proximity Sensing

Detects objects from a few centimeters up to about 10 cm away — the basis of touchless interaction with devices.

Gesture Detection

Recognizes up, down, left, and right swipes — control media or navigate menus without touching anything.

What the Sensors Can Do — Motion & Sound

Accelerometer

Measures acceleration along X, Y, and Z — motion, tilt, and free-fall detection in phones and wearables.

Gyroscope

Measures rotation rate around X, Y, and Z — stabilization in drones, gesture recognition, gaming controllers.

Microphone + FFT

The PDM mic converts sound into electrical signals; an FFT reveals the frequencies — plotted over time, that's a spectrogram.

How can we monitor streamed data?

- Open sketch in ‘**ble_dashboard**’:

[Google Drive → EE105 dashboard sketch (.ino)]

sketch

- Install the 5 required libraries

or

libraries

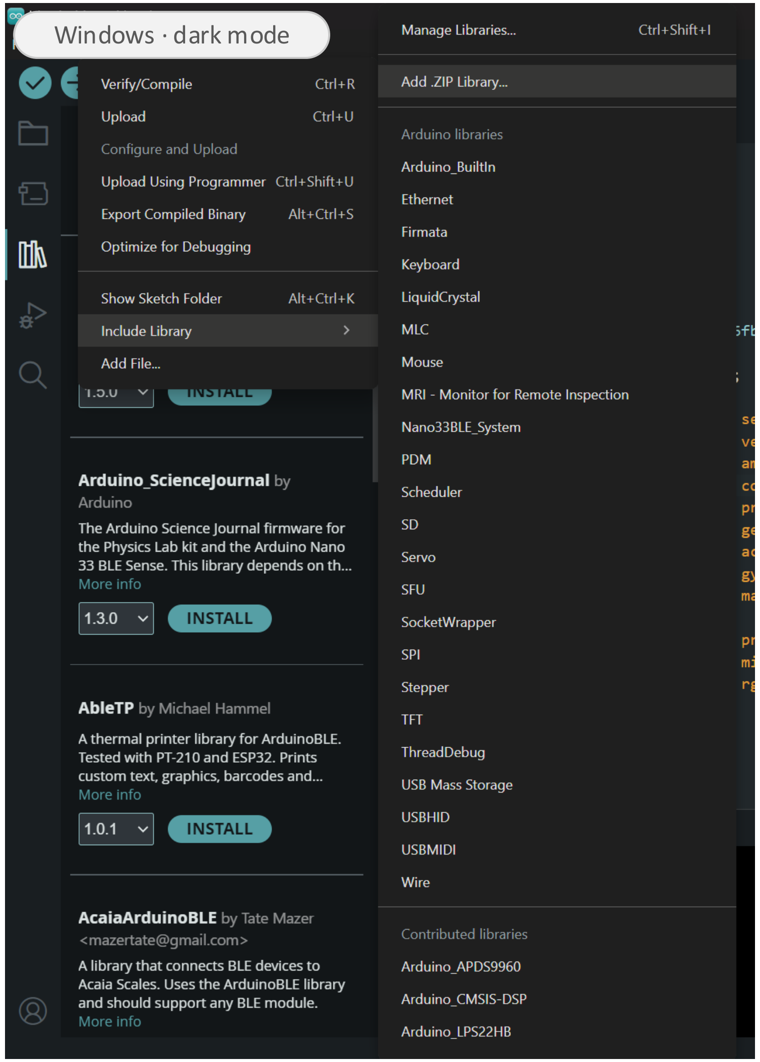
- Download the libraries (all 5) needed to run the above script:

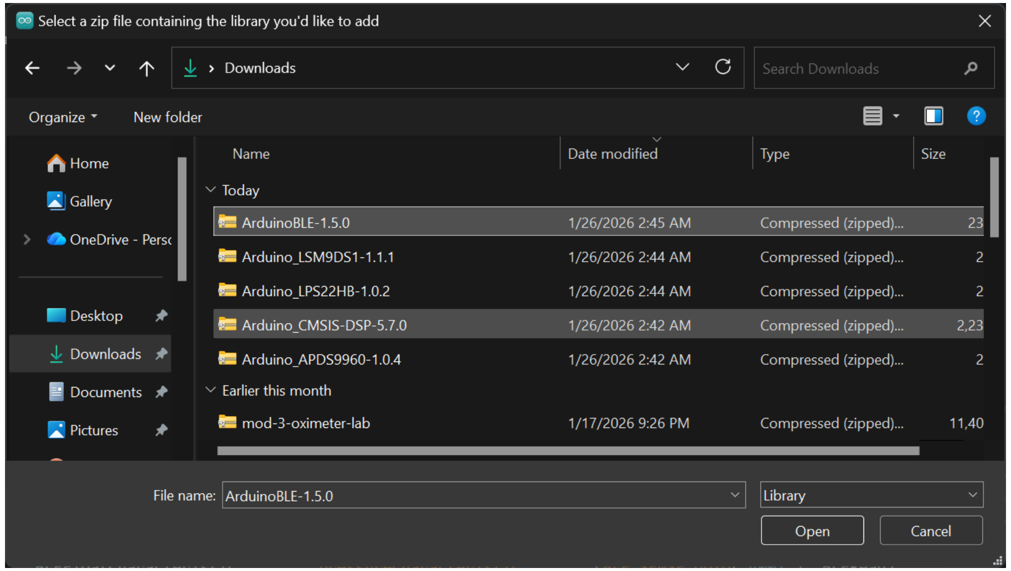
[Google Drive → required libraries (5 ZIP files)]

How can we monitor streamed data?

Installing Libraries:

- Sketch
- Include Library
- Add ZIP Library
- Repeat for each downloaded Library





How can we monitor streamed data?

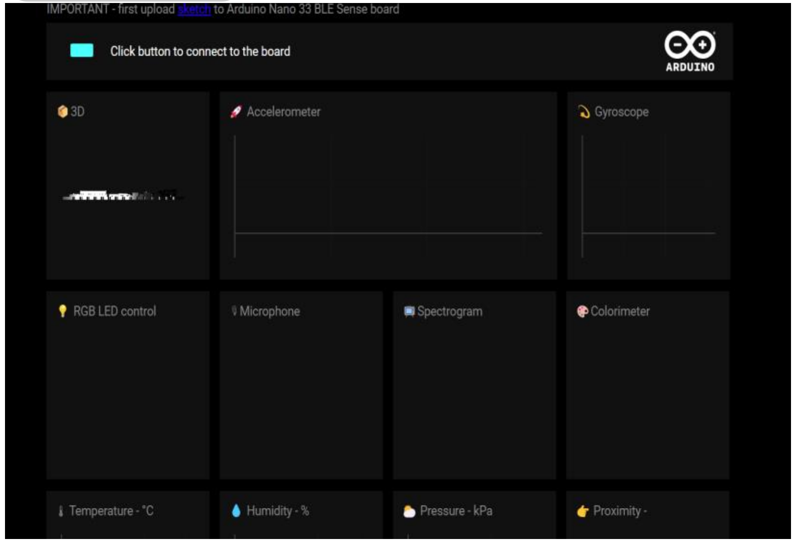
- Verify and upload
- Open the following link:
  [arduino.github.io → BLESense test dashboard]

Open the dashboard

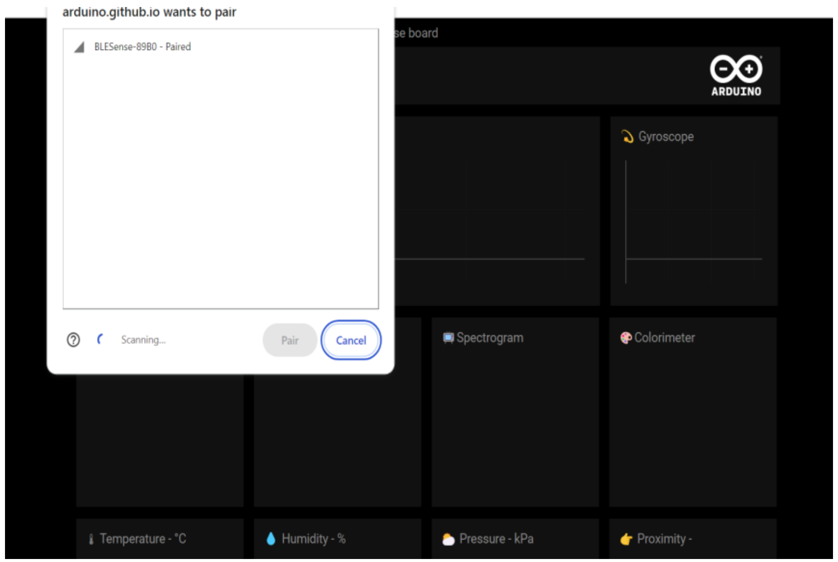
Pairing With the Dashboard

1.After uploading the code, click on the green button

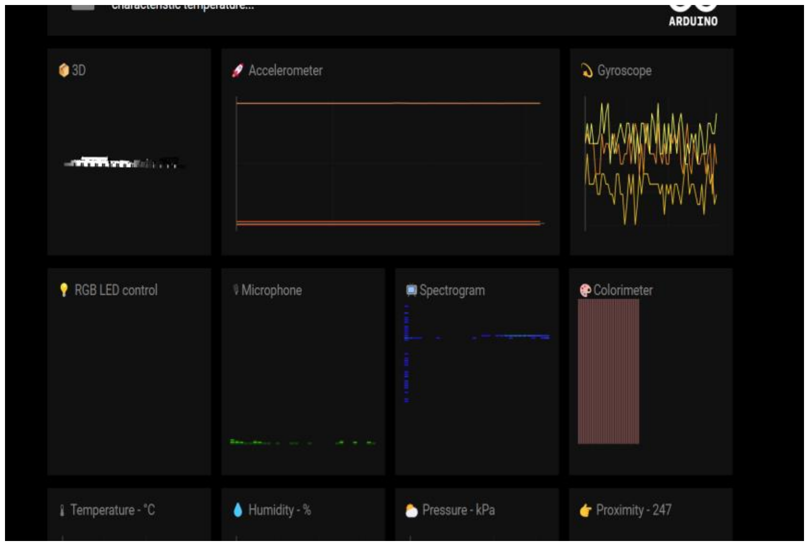
Google Chrome



2. Now your Arduino BLE will pair with the web app



3. The sensor data is streaming to the web app



Classwork 1

- **Solve all the tasks in the lab completion form.**

- Hint 1: This is how to define a function in the C programming language.

return type

Function Name

Parameters

Function header

```c
int  add ( int a,  int b)
{
    int sum  =  a + b;
    return ( sum );
}
```

no semicolon

Function body

Return statement

- Hint 2: In C / Arduino, you cannot use ^ for exponentiation. Instead, you must use the pow() function from <math.h>: pow(base, exponent)